LEARNER GUIDE

# Measuring and drawing angles

Angles Tell Us Why · Lesson 1. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

## What this lesson is about

An **angle** is an amount of **turn**, measured in **degrees** (°). A full turn is 360°, a right angle (a quarter turn) is 90° and a straight line (half a turn) is 180°. Angles less than 90° are **acute**, between 90° and 180° **obtuse**, and more than 180° **reflex**. The length of the arms does not change an angle. A protractor has two scales: estimate first, and the estimate tells you which one to read.

## What you are learning to notice

- An angle measures the turn between two arms, not the length of the arms.
- Turning the whole drawing does not change the angle.
- Acute: less than 90°. Right: 90°. Obtuse: between 90° and 180°. Reflex: between 180° and 360°.
- Read the protractor scale that starts at 0 on one arm.
- A reflex angle is 360° minus the turn the other way.

## Moving through it, one step at a time

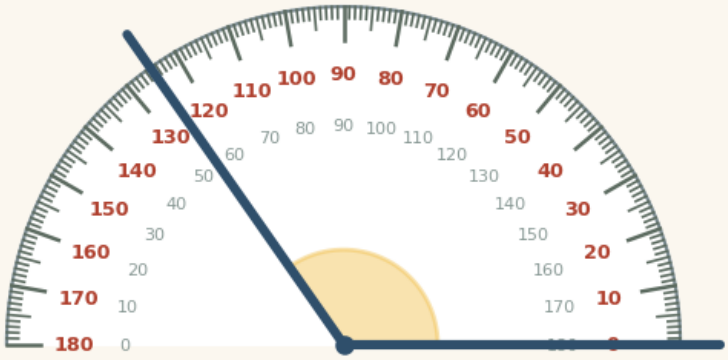
Estimate first: is the angle less than a right angle, between a right angle and a straight line, or more than a straight line? Put the protractor's centre on the vertex, and its 0 line along one arm. Read the scale that starts at 0 on that arm, where the other arm crosses it. Check the reading against your estimate. For a reflex angle, measure the turn the other way and take it from 360°.

## Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

**Example 1 — Measuring an obtuse angle**

How big is this angle?

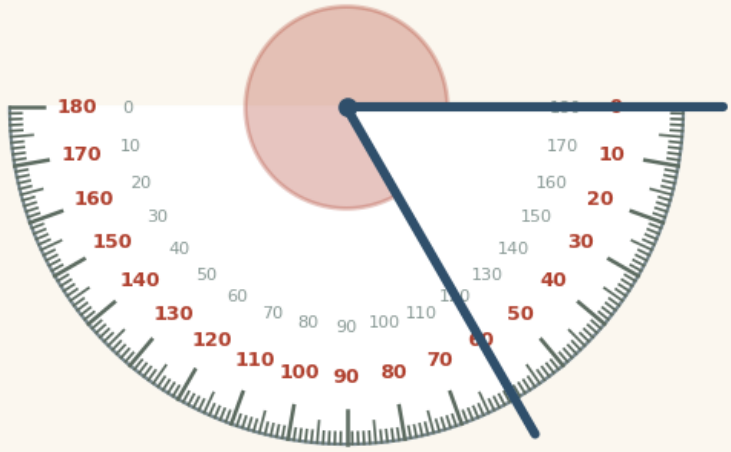


| | |
|---|---|
| **1.** Estimate: more than a right angle, less than a straight line. | obtuse |
| **2.** The red scale starts at 0 on the base arm. Where does the other arm cross it? | 125 |
| **3.** The grey scale shows 55 there, but 55° would be acute. | not 55 |
| **4.** Check against the estimate. | 125° is obtuse ✓ |

**The angle is 125°, an obtuse angle.**

**Example 2 — Measuring a reflex angle**

How big is the shaded reflex angle?



| | |
|---|---|
| **1.** Estimate: past a straight line, so | reflex |
| **2.** The protractor reaches only 180°. Flip it below the line and measure the turn the other way. | 60° |
| **3.** The two turns together make a full turn. | 360° |
| **4.** So the reflex angle is | 360° – 60° = 300° |

**The reflex angle is 300°.**

**A gentle reminder**

You can pause. You can return. You do not need to complete everything in one sitting.

## If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (right angles and quarter turns), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

**How the worked solution unlocks**

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

## Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

## Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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